AI505

Optimization

Second-Order Methods

Marco Chiarandini

Department of Mathematics & Computer Science
University of Southern Denmark

Outline

1. Newton Method

2. Secant Method

3. Quasi-Newton Method

Descent Direction Methods

How to select the descent direction?

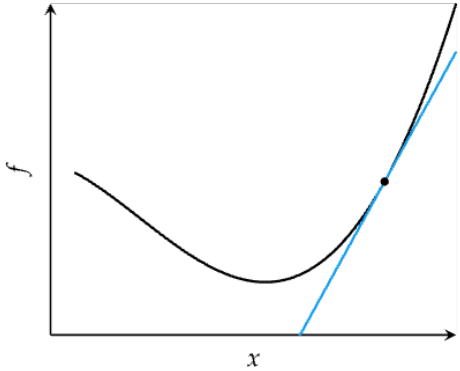
- first-order methods that rely on gradient
- second-order methods that rely on Hessian information

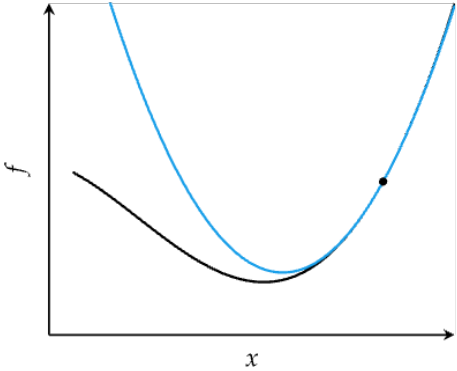
Advantages of second order methods in descent algorithms:

- way of accelerating the iteration [Davidon mid 1950s]
- additional information that can help improve the local model for informing the selection of
  - directions and
  - step lengths

Second-Order Methods

- Locally approximate function as quadratic
- Comparison of first-order and second order approximations





Outline

1. Newton Method

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3. Quasi-Newton Method

Newton's Method – Univariate

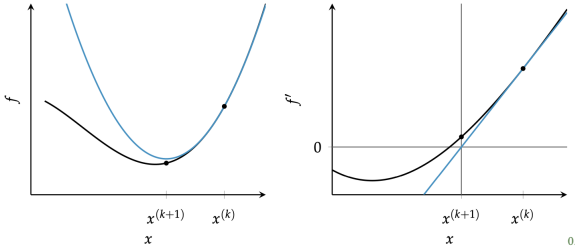
- Approximate a function using second-order Taylor series expansion
- analytically obtain the location where a quadratic approximation has a zero gradient.
- use that location as the next iteration to approach a local minimum.
- Univariate function

$$q(x) = f(x_k) + (x - x_k)f'(x_k) + \frac{(x - x_k)^2}{2} f''(x_k)$$

$$\frac{\mathrm{d}q(x)}{\mathrm{d}x} = f'(x_k) + (x - x_k)f''(x_k) = 0$$

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$\equiv$ finding roots of derivative function



Newton's Method - Multivariate

- Multivariate function

$$f(\mathbf{x}) \approx q(\mathbf{x}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T(\mathbf{x} - \mathbf{x}_k) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_k)^T H_k(\mathbf{x} - \mathbf{x}_k)$$

- H is the Hessian matrix
- Evaluate the gradient and set it to zero:

$$\nabla \mathbf{q}(x) = \nabla f(\mathbf{x}_k) + H(\mathbf{x}_k)(\mathbf{x} - \mathbf{x}_k) = 0$$

- Multivariate update rule

$$\boxed{\mathbf{x}_{k+1} = \mathbf{x}_k - H_k^{-1}\nabla f(\mathbf{x}_k)}$$

- (If f is quadratic and its Hessian is positive definite, then the update converges to the global minimum in one step.)

Algorithm

Input: $\nabla f, H, \mathbf{x}_0, \epsilon, k_{max}$
Output: $\mathbf{x}^*$
Set $k = 0, \Delta = 1, \mathbf{x} = \mathbf{x}_0$;
while $\|\Delta\| > \epsilon$ and $k \leq k_{max}$ **do**
  $\Delta = H(\mathbf{x})^{-1}\nabla f(\mathbf{x})$;
  $\mathbf{x} = \mathbf{x} - \Delta$;
  $k = k + 1$;

It can be modified to only give a descent direction $\mathbf{d} = -H(\mathbf{x})^{-1}\nabla f(\mathbf{x})$ and leave the step size to be determined with line search.

Newton's method – Example

Minimize **Booth's function**:

$$f(x) = (x_1 + 2x_2 - 7)^2 + (2x_1 + x_2 - 5)^2$$

- $\mathbf{x}_0 = [9, 8]$
- The gradient of Booth's function is:

$$\nabla f(x) = [10x_1 + 8x_2 - 34, 8x_1 + 10x_2 - 38]$$

- The Hessian of Booth's function is:

$$H(x) = \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix}$$

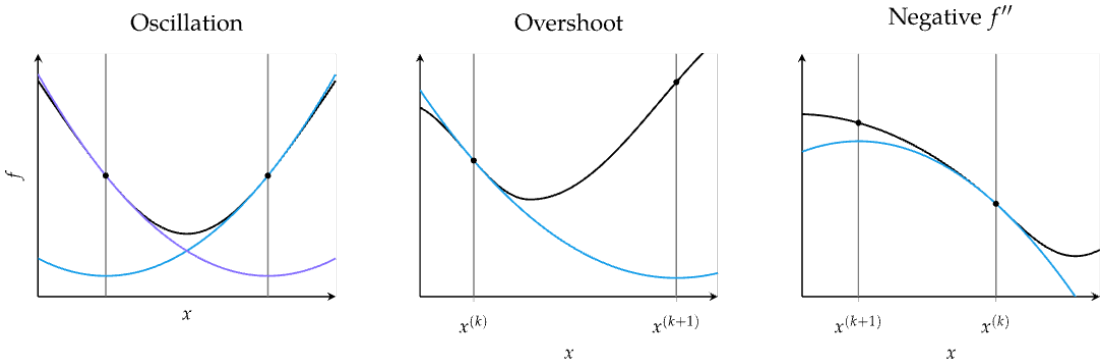
- The first iteration of Newton's method yields:

$$\begin{aligned}\mathbf{x}_1 &= \mathbf{x}_0 - H(\mathbf{x}_0)^{-1}\nabla f(\mathbf{x}_0)\\ &= \begin{bmatrix}9\\8\end{bmatrix} - \begin{bmatrix}10 & 8\\8 & 10\end{bmatrix}^{-1} \cdot \begin{bmatrix}10\cdot 9 + 8\cdot 8 - 34\\8\cdot 9 + 10\cdot 8 - 38\end{bmatrix} = \begin{bmatrix}9\\8\end{bmatrix} - \begin{bmatrix}10 & 8\\8 & 10\end{bmatrix}^{-1} \cdot \begin{bmatrix}120\\114\end{bmatrix} = \begin{bmatrix}1\\3\end{bmatrix}\end{aligned}$$

- Second iteration: The gradient at $\mathbf{x}_1$ is zero, so we have converged after a single iteration. The Hessian is positive definite everywhere, so $\mathbf{x}_1$ is the global minimum.

Newton's Method

Common causes of error in Newton's method



Strong Convexity

Recall that for a symmetric matrix A, $A \succeq 0$ means that A is positive semidefinite, i.e.

$$\mathbf{x}^\top A\mathbf{x} \geq 0 \quad \forall \mathbf{x} \in \mathbb{R}^n.$$

Hence, for a Q symmetric matrix, and I the identity matrix, the condition

$$Q \succeq cI \qquad \text{means:} \qquad Q - cI \succeq 0.$$

In scalar form:

$$\mathbf{x}^T(Q - cI)\mathbf{x} = \mathbf{x}^T Q\mathbf{x} - c\|\mathbf{x}\|^2 \geq 0.$$

$$\mathbf{x}^\top Q\mathbf{x} \geq c\|\mathbf{x}\|^2.$$

this was explained wrongly in class
and it has been corrected here

Strong Convexity

- Intuition: the second order derivative is positive, ie, $f''(x) \geq c > 0$.
- For high dimension: the Hessian matrix is positive definite.

Definition (Strongly Convex Function (Lower Curvature Bound))

The function $f : \mathbb{R}^d \to \mathbb{R}$ twice-differentiable function (ie, $f \in C^2(\mathbb{R}^n)$) is **c-strongly convex** if there exists a constant $c > 0$ such that for all $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^d$:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{c}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

or, equivalently, if there exists $c > 0$:

$$\nabla^2 f(\mathbf{x}) \succeq cI \qquad \text{for all } \mathbf{x} \in \mathbb{R}^d$$

When $c = 0$, we recover the basic inequality characterizing convexity; for $c > 0$ we obtain a better lower bound on $f(\mathbf{x})$ than follows from convexity alone.

Convex $\to$ no flat "bending downward"; Strongly convex $\to$ strictly curved upward everywhere

Strong Convexity

Implications:

- Uniqueness of Minimizer:
  f has **exactly one** minimizer, denoted as $\mathbf{x}^* \in \mathbb{R}^d$ with $f^* \stackrel{def}{=} f(\mathbf{x}^*)$.

- $f(x) - f^* \leq \frac{1}{2c}\|\nabla f(x)\|^2$ (strong inequality with $\mathbf{y} = \mathbf{x}^*$ and $\nabla f(\mathbf{y}) = 0$)
  quadratic growth: function grows at least like a quadratic bowl centered at $\mathbf{x}^*$

Strong Convexity

Interpretation

- Compared to ordinary convexity:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}),$$

  strong convexity adds a quadratic curvature term.
  It enforces that the function grows at least quadratically away from any point.
  Equivalently:

$$f(\mathbf{x}) - \frac{c}{2}\|\mathbf{x}\|^2 \text{ is convex.}$$

- Monotonicity of Gradient
  Strong convexity is equivalent to strong monotonicity of the gradient:

$$(\nabla f(\mathbf{x}) - \nabla f(\mathbf{y}))^\top (\mathbf{x} - \mathbf{y}) \geq c\|\mathbf{x} - \mathbf{y}\|^2 \quad \forall \mathbf{x}, \mathbf{y}.$$

  This is extremely important in convergence proofs.

Strong Convexity

Eigenvalue Interpretation:

Because Q is symmetric, it has an orthonormal eigenbasis.
Let its eigenvalues be $\lambda_1, \ldots, \lambda_n$.
Then:

$$Q \succeq cI \iff \lambda_i \geq c \quad \forall i.$$

In other words:

$$\lambda_{\min}(Q) \geq c.$$

So the smallest curvature direction of Q is at least c.

If

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix},$$

then eigenvalues are 1 and 100. So:
So:

- $Q \succeq I$
- $Q \not\succeq 2I$ (since $1 < 2$)

The smallest curvature direction determines the strong convexity constant.

Strong Convexity

Geometric Meaning:

Consider the quadratic function

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top Q\mathbf{x}.$$

Its curvature in direction $\mathbf{v}$ is $\mathbf{v}^\top Q\mathbf{v}$.
The condition $Q \succeq cI$ means:

$$\mathbf{v}^\top Q\mathbf{v} \geq c\|\mathbf{v}\|^2 \quad \forall \mathbf{v}.$$

So:
The function curves upward at least as much as a bowl with curvature c in every direction.

Smoothness Constant

- Definition in Optimization. It is NOT the smoothness in Mathematics (C^∞)
- Lipschitzness controls the changes in function value, while smoothness controls the changes in gradients.

Definition (Lipschitz Gradient (Upper Curvature Bound))

We say $f(\mathbf{x})$ is L-**smooth** when

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2, \quad L > 0$$

Some alternative definitions

1. $L\|x - x_0\|_2^2/2 - f(x)$ is convex (upper bounded by a quadratic function)
2. $-LI \preceq \nabla^2 f(x) \preceq LI$ (f has Lipschitz Hessian)
3. $\langle \nabla f(\mathbf{y}) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \leq L\|\mathbf{y} - \mathbf{x}\|^2$ (weaker when not require convexity)

Lipschitz Gradient

Definition (Lipschitz Gradient (Upper Curvature Bound))

f has an L-Lipschitz gradient if

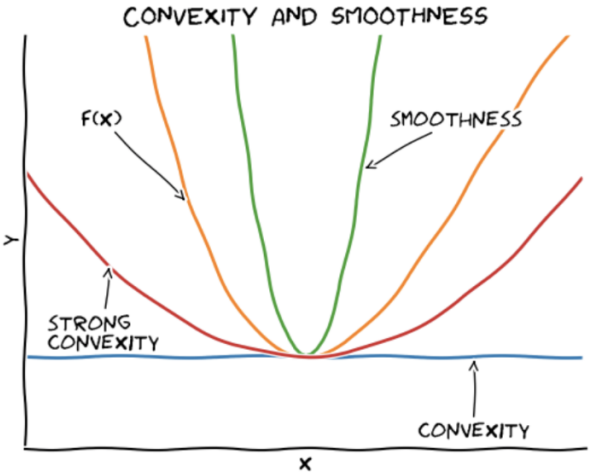
$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|.$$

Equivalent (if $f \in C^2(\mathbb{R}^n)$):

$$\nabla^2 f(\mathbf{x}) \preceq LI \qquad \text{for all } \mathbf{x} \in \mathbb{R}^n$$

This gives a uniform upper bound on curvature.

Illustration of Strong Convexity and Smoothness



[illustration by Zhanyu Wang, 2021]

<u>Combined Condition</u>:
If both hold:

$$cI \preceq \nabla^2 f(\mathbf{x}) \preceq LI \qquad \text{for all } \mathbf{x} \in \mathbb{R}^n$$

then all eigenvalues of the Hessian satisfy:

$$c \leq \lambda_i(\nabla^2 f(\mathbf{x})) \leq L.$$

So:
Strong convexity $\rightarrow$ prevents flatness
Lipschitz gradient $\rightarrow$ prevents excessive curvature

Stable Step Size

We can derive the stable step size for gradient descent for quadratic problems.
Consider the quadratic problem

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} - \mathbf{b}^\top \mathbf{x}, \qquad \text{where } Q = Q^\top \succ 0.$$

The gradient is $\nabla f(x) = Qx - \mathbf{b}$ and gradient descent with step size α:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha(Q\mathbf{x}_k - \mathbf{b}).$$

We can derive that the step size that stabilizes the iteration is

$$\boxed{0 < \alpha < \frac{2}{\lambda_{\max}(Q)}}$$

The largest eigenvalue corresponds to the direction with the largest curvature: $L = \lambda_{\max}(Q)$. Thus:

$$\boxed{0 < \alpha < \frac{2}{L}.}$$

If the step size is too large relative to that curvature,
the method overshoots and oscillates.

Convergence rate of gradient descent for quadratic problems:

$$\|x_k - x_*\| \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \|x_0 - x_*\|$$

where

$$\kappa = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)} = \frac{L}{c}$$

is the optimization condition number of Q.

Stable Step Size - Line Search

- If relevant parameters describing the regularity and curvature of the objective function (Lipschitz constant, smoothness constant, strong convexity parameter) are known they can be used to tune the algorithm's own parameters.

- These parameters are often unknown and difficult to estimate, and Line search is a powerful technique heavily used in practice.

Newton's Method - Multivariate

Theorem

If f is a strongly convex function with Lipschitz Hessian (aka, has $\nabla^2 f(\mathbf{x})$ that is Lipschitz continuous), then provided that x_0 is close enough to $x_* = \arg\min f(x)$, the sequence $x_0, x_1, x_2, \ldots$ generated by Newton's method will converge to the (necessarily unique) minimizer x_* of f quadratically fast. That is,

$$\|x_{k+1} - x_*\| \leq \frac{1}{2}\|x_k - x_*\|^2, \qquad \forall k \geq 0.$$

Lipschitz Hessian is a third order information on the variation of curvature.
Strong convexity says: The bowl never flattens. Lipschitz Hessian says: The bowl shape cannot change too abruptly.

Convergence of Newton's Method

Theorem (Convergence of Netwons Method)

Suppose that f is twice differentiable and that the Hessian $\nabla^2 f(\mathbf{x})$ is Lipschitz continuous with constant L in a neighborhood of a strict local minimizer $\mathbf{x}^*$. Consider the iteration $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{p}_k$, where $\mathbf{p}_k = -\nabla^2(\mathbf{x}_k)^{-1}\nabla f(\mathbf{x}_k)$. Then:

1. *if the starting point $\mathbf{x}_0$ is sufficiently close to $\mathbf{x}^*$, the sequence of iterates converges to $\mathbf{x}^*$;*
2. *the rate of convergence of $\{\mathbf{x}_k\}$ is quadratic; and*
3. *the sequence of gradient norms $\{\|\nabla f_k\|\}$ converges quadratically to zero.*

Quadratic convergence means that, when the difference is small, the difference between the minimum and the iterate is approximately squared with every iteration.

Newton's Method

Equivalently, from the text book:

- A quadratic rate of convergence holds for Newton's method starting from x_0 within an interval $I = [x^* - \delta, x^* + \delta]$, for a root x^*, if

  1. $f''(x) \neq 0$ for all points in I,
  2. $f'''(x)$ is continuous on I, and
  3. $\frac{1}{2}\left|\frac{f'''(x_0)}{f''(x_0)}\right| < L\left|\frac{f'''(x^*)}{f''(x^*)}\right|$ for some $L < \infty$
     sufficient closeness condition, ensuring that the function is sufficiently approximated by the Taylor expansion and no overshoot.

- Note that the Hessian matrix $\nabla^2 f(\mathbf{x}_k)$ may not always be positive definite, and the Netwon step: $\mathbf{d}_k = -\nabla^2(\mathbf{x}_k)^{-1}\nabla f(\mathbf{x}_k)$ not always be a descent direction. (There are approaches for obtaining a globally convergent iteration based on the Newton step: a line search approach, in which the Hessian $\nabla^2 f(\mathbf{x}_k)$ is modified, if necessary, to make it positive definite)

Outline

1. Newton Method

2. Secant Method

3. Quasi-Newton Method

Secant Method – Univariate

- For univariate functions, if the second derivative is unknown, it can be approximated using the secant method

$$f''(x_k) \approx \frac{f'(x_k) - f'(x_{k-1})}{x_k - x_{k-1}}$$

- Update equation

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f'(x_k) - f'(x_{k-1})} f'(x_k)$$

- It requires an additional initial design point and suffers from the same problems as Newton's method and may take more iterations to converge due to approximating the second derivative.

Outline

1. Newton Method

2. Secant Method

3. Quasi-Newton Method

Quasi-Netwon Methods – Multivariate

- Automatic differentiation tools may not be applicable in many situations, and it may be much more costly to work with second derivatives in automatic differentiation software than with the gradient.
- Quasi-Newton methods, like steepest descent, require only the gradient of the objective function to be supplied at each iterate.
- By measuring the changes in gradients, they construct a model of the objective function that is good enough to produce superlinear convergence.
- The improvement over steepest descent is dramatic, especially on difficult problems.

Quasi-Netwon Methods – Multivariate

Use an approximation $Q_k \approx H^{-1}(\mathbf{x}_k)$

Input: $\mathbf{x}_0$, convergence tolerance $\epsilon > 0$, Q_0 (typically the $n \times n$ identity matrix)
Output: x^*
Set $k \leftarrow 0$;
while $\|\nabla f(\mathbf{x}_k)\| > \epsilon$ **do**
  Compute search direction $\mathbf{d}(x_k) = -Q_k \nabla f(\mathbf{x}_k)$;
  Set $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}(\mathbf{x}_k)$ where α_k is computed from a line search procedure to satisfy the Wolfe conditions;
  Define $\boldsymbol{\delta}_{k+1} \stackrel{def}{=} \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\boldsymbol{\gamma}_{k+1} \stackrel{def}{=} \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$;
  Compute Q_{k+1};
  $k \leftarrow k + 1$;

- Davidon-Fletcher-Powell (DFP) method
- Broyden-Fletcher-Goldfarb-Shanno (BFGS) method
- Limited-memory BFGS (L-BFGS) method

Davidon-Fletcher-Powell (DFP) method

$$Q_{k+1} = Q_k - \frac{Q_k \gamma_k \gamma_k^T Q_k}{\gamma_k^T Q_k \gamma_k} + \frac{\delta_k \delta_k^T}{\delta_k^T \gamma_k}$$

where all terms on the right hand side are evaluated at the same iteration k.

The update for Q in the DFP method has three properties:

- Q remains symmetric and positive definite.
- If $f(x) = \frac{1}{2}\mathbf{x}^T A\mathbf{x} + \mathbf{b}^T\mathbf{x} + c$, then $Q = A^{-1}$. Thus the DFP has the same convergence properties as the conjugate gradient method.
- For high-dimensional problems, storing and updating Q can be significant compared to other methods like the conjugate gradient method.

Broyden-Fletcher-Goldfarb-Shanno (BFGS) method

$$Q_{k+1} = Q_k - \left(\frac{\boldsymbol{\delta}_k \boldsymbol{\gamma}_k^T Q_k + Q_k \boldsymbol{\gamma}_k \boldsymbol{\delta}_k^T}{\boldsymbol{\delta}_k^T \boldsymbol{\gamma}_k} \right) + \left(1 + \frac{\boldsymbol{\gamma}_k^T Q_k \boldsymbol{\gamma}_k}{\boldsymbol{\delta}_k^T \boldsymbol{\gamma}_k} \right) \frac{\boldsymbol{\delta}_k \boldsymbol{\delta}_k^T}{\boldsymbol{\delta}_k^T \boldsymbol{\gamma}_k}$$

BFGS better than DFP with approximate line search but still uses an $n \times n$ dense matrix.

<u>Theorem</u>: Suppose that f is twice continuously differentiable and that the iterates generated by the BFGS algorithm converge to a minimizer $\mathbf{x}^*$ at which the Hessian matrix G is Lipschitz continuous Suppose also that the sequence $\|\mathbf{x}_k - \mathbf{x}^*\|$ converges to zero rapidly enough that $\sum_{k=1}^{\infty} \|\mathbf{x}_k - \mathbf{x}^*\| < \infty$. Then $\mathbf{x}_k$ converges to $\mathbf{x}^*$ at a superlinear rate (ie, faster than linear).

Limited-memory BFGS (L-BFGS) method

For **large-scale unconstrained optimization**
It stores the last m values for δ and γ rather than the full inverse Hessian ($i = 1$ oldest, $i = m$ last).
Compute $\mathbf{d}$ at $\mathbf{x}$ as $\mathbf{d} = -\mathbf{z}_m$ using:

$$\mathbf{q}_m = \nabla f(\mathbf{x}_k) \qquad \mathbf{q}_i = \mathbf{q}_{i+1} - \frac{\boldsymbol{\delta}_{i+1}^T \mathbf{q}_{i+1}}{\boldsymbol{\gamma}_{i+1}^T \boldsymbol{\delta}_{i+1}} \boldsymbol{\gamma}_{i+1}, \quad i = m-1, \ldots, 1$$

$$\mathbf{z}_0 = \frac{\boldsymbol{\delta}_m \odot \boldsymbol{\delta}_m \odot \mathbf{q}_m}{\boldsymbol{\gamma}_m^T \boldsymbol{\gamma}_m} \qquad \mathbf{z}_i = \mathbf{z}_{i-1} + \boldsymbol{\delta}_{i-1} \left(\frac{\boldsymbol{\delta}_{i-1}^T \mathbf{q}_{i-1}}{\boldsymbol{\gamma}_{i-1}^T \boldsymbol{\delta}_{i-1}} - \frac{\boldsymbol{\gamma}_{i-1}^T \mathbf{z}_{i-1}}{\boldsymbol{\gamma}_{i-1}^T \boldsymbol{\gamma}_{i-1}} \right), \quad i = 1, \ldots, m$$

For minimization, the inverse Hessian Q must remain positive definite.
The initial Hessian is often set to the diagonal of

$$Q_0 = \frac{\boldsymbol{\gamma}_0 \boldsymbol{\delta}_0^T}{\boldsymbol{\gamma}_0^T \boldsymbol{\gamma}_0}$$

Computing the diagonal for the above expression and substituting the result into $\mathbf{z}_0 = Q_0 \mathbf{q}_0$ results in the equation for $\mathbf{z}_0$.

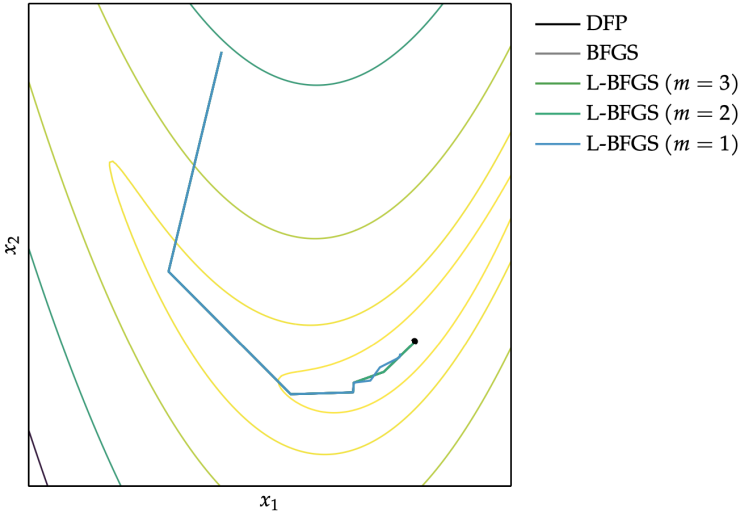
L-BFGS

(L-BFGS two-loop recursion)

Input:
Output: $Q_k \nabla f_k = \mathbf{z}$
Set $\mathbf{q} \leftarrow \nabla f_k$;
for $i = k-1, k-2, \ldots, k-m$ **do**
  $\alpha_i \leftarrow \frac{\delta_i^T \mathbf{q}}{\gamma_i \cdot \delta_i^T}$;
  $\mathbf{q} \leftarrow \mathbf{q} - \alpha_i \cdot \gamma_i$;
$\mathbf{z} \leftarrow Q_0 \mathbf{q}$;
for $i = k-m, k-m+1, \ldots, k-1$ **do**
  $\beta \leftarrow \frac{\gamma_i^T \cdot \mathbf{r}}{\gamma_i \cdot \delta_i^T}$;
  $\mathbf{z} \leftarrow \mathbf{z} + \delta_i(\mathbf{z}_i - \beta)$;

Input:
Output: x^*
Choose starting point $\mathbf{x}_0$, integer $m > 0$;
$k \leftarrow 0$;
while not convergence **do**
  Set Q_0;
  Compute $\mathbf{d}_k \leftarrow -Q_k \nabla f_k$ from Algorithm on the left;
  Compute $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k + \alpha_k \mathbf{d}_k$, where α_k is chosen to satisfy the Wolfe conditions;
  if $k > m$ **then**
    Discard the vector pair $\{\boldsymbol{\delta}_{k-m}, \gamma_{k-m}\}$ from storage;
  Compute and save $\boldsymbol{\delta}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$, $\gamma_k = \nabla f_{k+1} - \nabla f_k$;
  $k \leftarrow k+1$;

BFGS Methods - Comparison



Summary

- Incorporating second-order information in descent methods often speeds convergence.
- Newton's method is a root-finding method that leverages second-order information to quickly descend to a local minimum.
- The secant method and quasi-Newton methods approximate Newton's method when the second-order information is not directly available.
- In Python, methods implemented in the module scipy
  https://docs.scipy.org/doc/scipy/tutorial/optimize.html